# Wenyan Hu

Senior Spatial Analyst

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## About me

I am a GIS specialist with a PhD in Geomatic Engineering and three years of experience in the Victorian Public Service, focusing on enterprise GIS administration, spatial data governance, and geospatial modelling. I am proficient in ArcGIS Suite, Python, JavaScript, FME, C# and SQL, with experience developing web GIS applications, ArcGIS Pro add-ins, and automated geoprocessing workflows.

I am passionate about scalable geospatial solutions, data quality, and analytical tools that enhance decision-making, and I thrive in fast-paced, collaborative environments.

## Technical skills

### Programming

Python | JavaScript | TypeScript | C# | Java | SQL

### Web Dev

AWS | Node.js | ArcGIS Maps SDK for Javascript | Geoprocessing API

### Libraries and Tools

Pandas | Numpy | Arcpy | ArcGIS | PyTorch | TensorFlow | GeoPandas | MATSim

### Geospatial

ArcGIS Pro | ArcGIS Online | ArcGIS Enterprise | ArcGIS Survey123 | FME | QGIS | ENVI

## Languages

English
Chinese

## Experience

**Senior Spatial Analyst @ Department of Transport and Planning VIC** *2023 — Current*

Responsible for developing and maintaining **enterprise GIS applications**, **geoprocessing tools**, and **spatial data systems** to support planning analysis and decision-making.

**Teaching Assistant @ University of Melbourne** *2019 — 2024*

Tutor of underground and master level geospatial subjects on **Geographic Information System**, **Geospatial Informatics**, **Engineering Modelling**.

## Projects

**ArcGIS Enterprise Deployment and Development** *2025 — Current*

Deployed **ArcGIS Enterprise** on **AWS EC2** including **ArcGIS Server**, **Portal**, **Data Store**, and **Web Adaptor**. Managed **security groups**, migrated **spatial datasets**, and published **feature** and **Geoprocessing services**.

**Add-in Tools for Amendment and Gazetted Map Printing** *2024 — Current*

Migrated and enhanced **C# ArcGIS add-ins** to automate **planning scheme amendment** workflows, generating **amendment maps** and **gazetted outputs** for statutory planning documentation.

**Renewables WebApp and Heritage WebApp** *2024 — Current*

Developed and maintained **GIS web applications** for renewable and heritage data, supporting **ETL workflows**, **web map configuration**, and **user management**.

**Automated Spatial Data Sync and Map Services Publishing** *2024 — Current*

Built automated **Python workflows** to synchronize spatial data between **ShareDrive** and **AWS EC2**. Implemented **Git-based pipelines** for publishing and maintaining **ArcGIS Server map services**.

**Vic3D WebApp** *2023 — 2024*

Enhanced **3D web mapping platform** functionality including **layer management**, **user administration**, and **interface optimisation**.

**VicPlan – Definition Plan Tool** *2023 — 2024*

Migrated and enhanced **Definition Plan geoprocessing tools** to **ArcGIS Pro 3.x** and supported mapping teams with **tool development** and **spatial data management**.

**Urban Traffic Simulation Research – Melbourne** *2019 — 2021*

Performed **data processing** and **agent-based traffic simulation** to analyse dynamic road events, contributing to **peer-reviewed publications**.

## Education

*2019 — 2023* **University of Melbourne** Melbourne, Australia

Doctor of Philosophy, Engineering — Thesis: Information sensing, transmission and management between connected vehicles

*2016 — 2019* **East China Normal University** Shanghai, China

Master of Science, Physical Geography

*2012 — 2016* **Northwest A&F University** Shaanxi, China

Bachelor of Science, GIS